# GROUP 2

**1. Discuss the Application of Big Data Across Various Industries. ADITYA**

**1. Retail Industry**

**Retailers analyze customer purchase history and preferences.**

**Personalized product recommendations help improve cross-selling and up-selling.**

**Large retailers process huge volumes of customer transactions in real time.**

**Techniques such as customer churn analysis, market basket analysis, and collaborative filtering help understand customer buying patterns.**

**Big Data helps improve customer engagement and reduce operational costs.**

**2. E-Commerce and Online Marketing (ClickStream Analytics)**

**E-commerce companies analyze ClickStream data, which records users' clicks and browsing activities.**

**It helps businesses understand customer behavior and purchasing patterns.**

**ClickStream data can be combined with CRM data, such as demographics, sales history, and campaign information.**

**Companies can store historical browsing data for long-term trend analysis.**

**The data can be organized by user sessions and analyzed using visualization and analytics tools.**

**3. Banking and Financial Services**

**Big Data helps detect financial fraud by identifying suspicious transaction patterns.**

**Banks analyze large volumes of transactional and customer data.**

**Advanced analytics improves risk assessment and management.**

**It helps provide better customer service.**

**Real-time data analysis supports faster and better decision-making.**

**4. Healthcare Sector**

**Big Data supports data-driven decision-making in hospitals and healthcare systems.**

**It helps reduce costs and unnecessary expenditure.**

**Patient health records are stored and analyzed to improve treatment outcomes.**

**Healthcare data supports patient history tracking and medical research.**

**The sector has significant potential as more clinical and patient information becomes digitized.**

**5. Social Media and Web-Based Businesses**

**Social media platforms generate enormous amounts of data every day.**

**Data from platforms such as Facebook, LinkedIn, YouTube, and Instagram can be analyzed by businesses.**

**Big Data is used for sentiment analysis and brand monitoring.**

**It supports targeted advertising and personalized marketing.**

**Businesses can understand public opinion and trends in real time.**

**6. Telecommunications, Transport, and Entertainment**

**Transport companies use analytics for logistics optimization, route planning, and customer service.**

**Telecommunication companies use Big Data to monitor network performance and detect fraud.**

**Telecom providers can personalize services and offers for customers.**

**Entertainment companies analyze audience preferences and viewing behavior.**

**Big Data helps provide personalized content recommendations and supports content investment decisions.**

**7. Manufacturing**

**Big Data helps improve production planning and quality control.**

**Manufacturers use data analytics for demand forecasting.**

**Techniques such as regression-based forecasting help predict future trends.**

**Customer and market data help manufacturers respond quickly to changing demands.**

**Analytics can improve operational efficiency and reduce production costs.**

**8. Scientific Research and Government/Public Sector**

**Massive datasets are used for weather prediction and climate analysis.**

**Big Data supports oil exploration and seismic activity monitoring.**

**Governments use data for census analysis, policy-making, and economic planning.**

**Public and regulatory data supports large-scale research.**

**Open web resources and government datasets provide valuable information for researchers and decision-makers.**

**9. Supporting Data Sources Across Industries**

**Organizations use both internal and external data sources.**

**Internal sources include databases, business archives, and organizational documents.**

**External sources include public web data and social media.**

**Sensor data from vehicles, smart meters, and buildings provides real-time information.**

**Machine logs and enterprise systems such as ERP, CRM, and HR platforms generate valuable business data.**

**Combining structured, semi-structured, and unstructured data helps organizations gain a complete view of customers and operations.**

**2. Explain the various categories of data analytics and illustrate how they progress in terms of difficulty. AKANKSHA**

Analytics can be classified into two schools of thought:

Basic, Operationalized, Advanced, and Monetized Analytics

Analytics 1.0, Analytics 2.0, and Analytics 3.0

**1. First School of Thought:**

**1. Basic Analytics**

**This primarily is slicing and dicing of data to help with basic business insights.**

**It is about reporting on historical data.**

**It also includes basic visualization.**

**Difficulty: Low – because it mainly deals with basic insights from historical data.**

**2. Operationalized Analytics**

**It is operationalized analytics if it gets woven into the enterprise's business processes.**

**Difficulty: Moderate – because analytics is integrated into the enterprise's business processes.**

**3. Advanced Analytics**

**This largely is about forecasting for the future.**

**It uses predictive and prescriptive modeling.**

**Difficulty: High – because it involves forecasting and predictive/prescriptive modeling.**

**4. Monetized Analytics**

**This is analytics in use to derive direct business revenue.**

**Difficulty: Very High – because analytics is used to derive direct business revenue.**

**2. Second School of Thought**

| Analytics 1.0 | Analytics 2.0 | Analytics 3.0 |
|---|---|---|
| **Era: mid 1950s to 2009** | **2005 to 2012** | **2012 to present** |
| Descriptive statistics (report on events, occurrences, etc. of the past) | Descriptive statistics + predictive statistics (use data from the past to make predictions for the future) | Descriptive + predictive + prescriptive statistics (use data from the past to make prophecies for the future and at the same time make recommendations to leverage the situation to one's advantage) |
| Key questions asked: What happened? Why did it happen? | Key questions asked: What will happen? Why will it happen? | Key questions asked: What will happen? When will it happen? Why will it happen? What should be the action taken to take advantage of what will happen? |
| Data from legacy systems, ERP, CRM, and 3rd party applications. | Big data | A blend of big data and data from legacy systems, ERP, CRM, and 3rd party applications. |
| Small and structured data sources. Data stored in enterprise data warehouses or data marts. | Big data is being taken up seriously. Data is mainly unstructured, arriving at a much higher pace. This fast flow of data entailed that the influx of big volume data had to be stored and processed rapidly, often on massive parallel servers running Hadoop. | A blend of big data and traditional analytics to yield insights and offerings with speed and impact. |
| Data was internally sourced. | Data was often externally sourced. | Data is both being internally and externally sourced. |
| Relational databases | Database appliances, Hadoop clusters, SQL to Hadoop environments, etc. | In memory analytics, in database processing, agile analytical methods, machine learning techniques, etc. |

3. Differentiate between parallel and distributed systems. AKSHATHA

| PARALLEL COMPUTING | DISTRIBUTED COMPUTING |
|---|---|
| A parallel system uses **multiple processors** working together to perform a task. | A distributed system consists of multiple **independent machines/computers**. |
| The system is generally **tightly coupled**. | The system is generally **loosely coupled**. |
| Processors **co-operate** for query processing. | Each machine can run its **own individual applicati** |
| Processors may have access to **common memory**. | Each machine has its **own processor and memory** |
| Processors can also communicate using **message passing**. | The machines communicate with each other through a **network**. |
| The user is generally **unaware** of the parallelism taking place. | The machines can serve their respective users **independently**. |
| A number of processors work together as a **single closely connected system**. | A number of machines may be connected togethe to answer a **user query**. |
| Parallel processing divides work among processors to achieve **faster processing**. | Distributed processing divides **work/resources** among different networked machines. |
| **Example:** Multiple processors accessing a shared memory. | **Example:** Multiple computers, each with its own memory, connected through a network. |
| Failure of one processor typically **does not stop the entire system.** | Failure of one machine **may affect the whole syst** or **reduce overall performance**. |

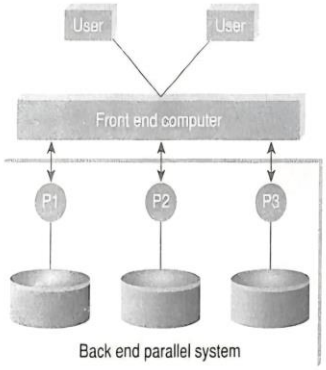



Figure 3.10 Parallel system.

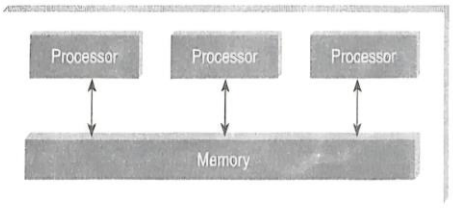



Figure 3.11 Parallel system.

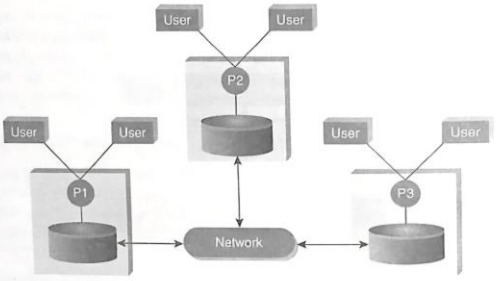



Figure 3.12 Distributed system.

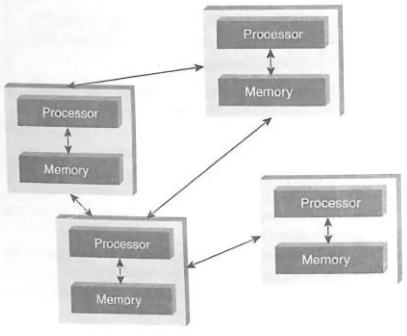



Figure 3.13 Distributed system.

4. Explain the following: DATTATREYA

i)SMP

ii)Shared-nothing architecture

**i) SMP – Symmetric Multiprocessing**

- **SMP stands for Symmetric Multiprocessing.**
- **SMP is a multiprocessor system in which two or more identical processors share a single main memory.**
- **All processors have equal access to shared memory and I/O devices.**
- **All processors are controlled by a single operating system (OS).**
- **Each processor can execute programs independently and work on different tasks at the same time.**
- **SMP is a tightly coupled multiprocessing system.**
- **A cache is usually associated with each processor to improve performance.**
- **A bus or interconnection network connects the processors, memory, and I/O devices.**
- **SMP is also called a shared-memory system.**
- **Processors can share workload dynamically, allowing tasks to be distributed among available processors.**
- **It provides efficient utilization of system resources because processors can access the same memory and devices.**
- **SMP is commonly used in servers, workstations, and high-performance computer systems.**

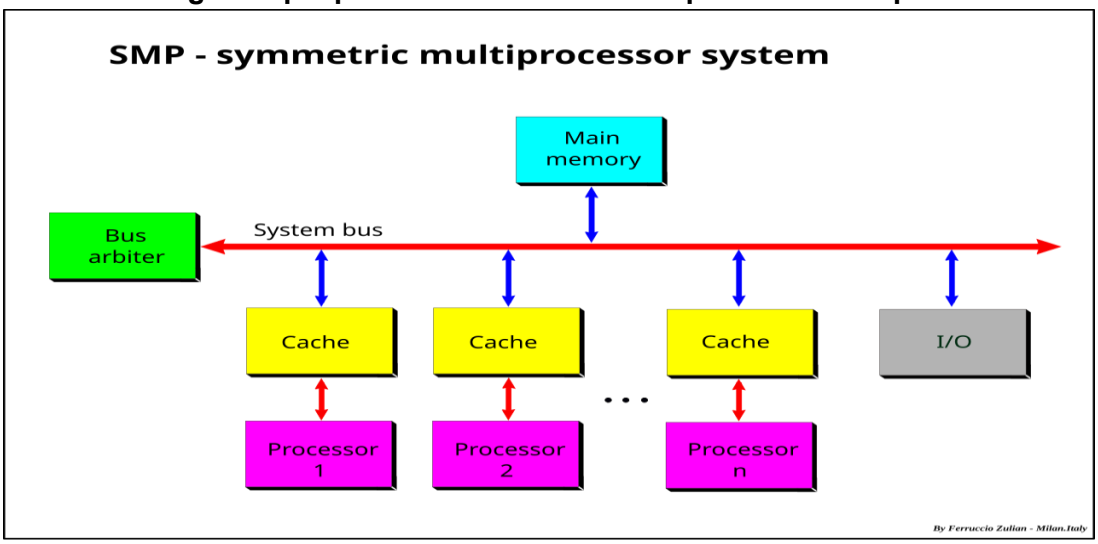
**Advantages of SMP**

1. **Faster Processing:**
   **– Multiple processors can execute tasks simultaneously.**
2. **Resource Sharing:**
   **– Processors share the same memory and I/O resources.**
3. **Reliability:**
   **– If one processor fails, other processors may continue working.**
4. **Better Performance:**
   **– Work can be divided among multiple processors.**
5. **Load Balancing:**
   **– Workload can be distributed among processors to avoid overloading one processor.**
6. **Efficient Resource Utilization:**
   **– Shared resources can be used efficiently by all processors.**
7. **Scalability:**
   **– Additional processors can be added to improve processing capacity, within system limits.**

**Disadvantages of SMP**

1. **Shared Memory Bottleneck:**
   **– Multiple processors accessing the same memory can cause delays and reduce performance.**
2. **Limited Scalability:**
   **– Adding too many processors may not provide proportional performance improvement because shared resources become overloaded.**
3. **Higher Cost:**
   **– SMP systems require specialized hardware and multiple processors, which can increase the cost.**
4. **Complex Operating System:**
   **– The operating system must manage multiple processors and coordinate their activities.**

5. **Synchronization Problems:**
   **– Processors working on shared data need synchronization, which can make programming more difficult.**
6. **Single Point of Failure:**
   **– Failure of shared memory or other major shared components can affect the entire system.**
7. **Heat and Power Consumption:**
   **– Using multiple processors can increase power consumption and heat generation.**




**ii)Shared-Nothing Architecture**

There are **three most common types of architecture** for multiprocessor high-transaction-rate systems:

**1. Shared Memory (SM)**

- A common central memory is shared by multiple processors.
- All processors can access the shared memory.
- **Example:** SMP (Symmetric Multiprocessing).

**2. Shared Disk (SD)**

- Multiple processors have their own private memory.
- They share a common collection of disks.
- Each processor can access the shared disk/storage.

**3. Shared Nothing (SN)**

- Each processor/node has its own **private memory and storage**.
- There are **no shared resources** between the nodes.
- Nodes communicate with each other through a **network**.

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**Shared-Nothing Architecture**

**Definition:**

Shared-Nothing Architecture is a distributed computing architecture in which each node works independently with its own **CPU, memory, and storage**.

Shared-Nothing Architecture was proposed by **Michael Stonebraker in 1986**.

**Features**

**1. Independent Resources**

Each node has its own **CPU, memory, and storage**, allowing it to work independently.

**2. No Shared Resources**

Nodes do not share **memory or storage** with other nodes.

**3. Network Communication**

Nodes communicate with each other through a **network** to exchange data and coordinate tasks.

**4. Independent Operation**

Each node can **process tasks independently** without depending completely on other nodes.

**5. Horizontal Scaling**

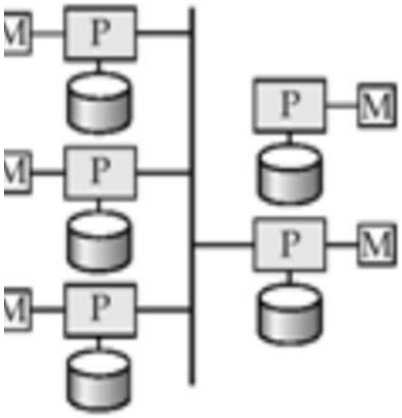
New nodes can be added easily to **increase the capacity and performance** of the system.

**6. Fault Tolerance**

If one node fails, other nodes can **continue working**, improving system availability.

**7. Cost Effectiveness**

The system can use multiple **smaller and less expensive machines**, making it more cost-effective.




(c) shared nothing

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**Advantages**

**1. Eliminates Single Point of Failure (SPOF)**

If one node fails, the other nodes can **continue working**, so the entire system does not necessarily stop.

**2. Supports Non-Disruptive Upgrades**

Individual nodes can be **upgraded or maintained without stopping the entire system**.

**3. Allows Easy Expansion**

New nodes can be added to **increase the system's capacity and performance**.

**4. Provides High Availability**

The system can remain available because **other nodes can continue working** even when one node fails.

**5. Supports Self-Healing Capabilities**

The system can detect node failures and **redistribute tasks to healthy nodes** to maintain operation.

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**Disadvantages**

**1. Hard to Program**

Applications are more difficult to develop because they must **manage data distribution and communication between nodes**.

**2. Adding New Nodes May Require Reorganization**

When new nodes are added, **data and workloads may need to be redistributed** among the nodes.

**3. Managing Communication Between Nodes Can Be Complex**

Nodes communicate through a network, so **coordinating data and tasks between different nodes can be difficult**.

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**Example**

A **large e-commerce platform** can distribute transactions among multiple servers. If one server fails, the other servers can **continue processing transactions**, keeping the system available.

5.Explain CAP Theorem. DEEKSHA

In a Distributed System, data is stored and accessed across multiple servers or nodes. Maintaining reliability, performance, and consistency across all these nodes is challenging.
To address this problem, Eric Brewer introduced the CAP Theorem in 2000. It is also called Brewer's Theorem.

The CAP theoretn is also called the Brewer's Theorem. It States that in a distributed computing environment (a collection Of interconnected nodes Chat share data), it is impossible to provide the following guarantees.
At best you can have two Of the following three — one must be sacrificed.
I. Consistency
2. Availabilitv
3. Parcition tolerance

2. CAP stands for

C – Consistency
A – Availability
P – Partition Tolerance

The CAP theorem states:

> A distributed system cannot guarantee all three properties—Consistency, Availability, and Partition Tolerance—at the same time. At most, two can be fully guaranteed.

3. Components of CAP Theorem

C – Consistency

Consistency means that all nodes return the most recent and correct data.

Every read receives the latest write.

All users see the same data at the same time.

Example: If a Facebook status is updated, users should see the updated status.

A – Availability

Availability means that every request receives a response, even if the response may not contain the most recent data.

The system remains operational.

Users can access the system.

However, the returned data may sometimes be stale.

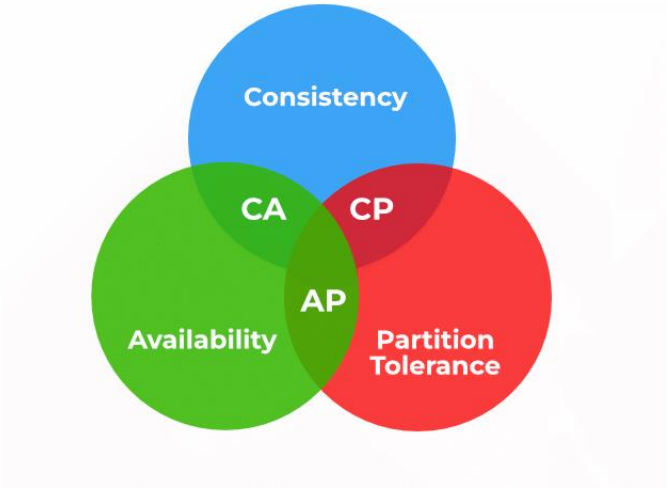
P – Partition Tolerance

Partition tolerance means that the system continues to operate even when communication between some nodes is interrupted.

A network failure may divide the system into separate groups.

The system continues functioning despite this network partition.

This is especially important in distributed systems.




4. CAP Triangle

The three properties can be represented as:

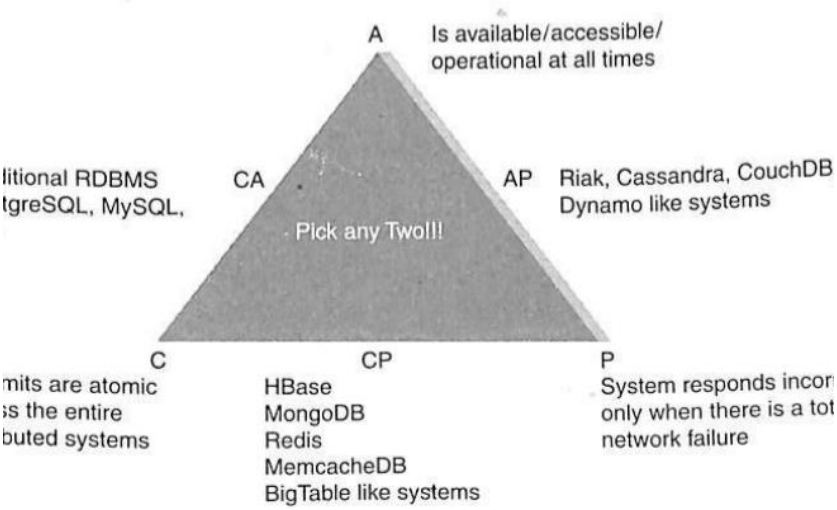



**Figure 3.15** Databases and CAP.

A distributed system can generally guarantee only two of the three properties.

5. Three CAP Combinations

CA – Consistency + Availability

The system provides consistent and available data.

However, it cannot tolerate a network partition.

Suitable when network failures/partitions are not expected or can be avoided.

CP – Consistency + Partition Tolerance

The system continues operating during a network partition.

It prioritizes correct and consistent data.

Some requests may become unavailable until consistency can be maintained.

AP – Availability + Partition Tolerance

The system continues responding even during a network partition.

It prioritizes availability.

Some returned data may be temporarily inconsistent or stale.

6. Example

Consider a social media application where a user updates their status.

Suppose the application has multiple servers and a network failure separates some servers.

CA: The system provides consistent and available data, but cannot handle the partition.

CP: The system maintains correct data during the partition, but some users may temporarily be unable to access the data.

AP: The system remains available, but some users may temporarily see an older version of the status.